

EDUCATION

University Of California, Berkeley 2017 - 2022
M.S. + Ph.D. in Engineering

- Minors in Mathematics and Computer Science
- Recipient of the IEOR PhD Fellowship (Research Award, 2017)

Moscow Institute of Physics And Technology 2013 - 2017
B.S. in Applied Math, Computer Science and Physics

- Graduated summa cum laude, GPA 3.98

WORK EXPERIENCE

University of Hawai'i at Manoa 2023 - current
Assistant Professor

- Founded and direct the Hawai'i Artificial Intelligence Initiative (HawAII), managing three teams of 3-6 graduate students working on different layers of foundational model research and deployment:
 - *Biotech*: Pre-training large-scale foundational models for biology and healthcare applications in collaboration with the UH Cancer Center and School of Medicine.
 - *Robotics*: Building embodied AI and robust control systems for the autonomous maintenance and operation of safety-critical energy infrastructure.
 - *Agents*: Constructing cognitive agentic architectures and deploying an AI educational platform (Alu) across university courses.
- Teach graduate-level courses: ECE 605 (Large-Scale AI Development and Deployment), ECE 491B/693B (Large Language Models), EE 417/617 (Optimization and Control); and undergraduate courses: ECE 405 (Introduction to Large-Scale AI Systems).
- Founded and advise the Foundational Models for Robotics (HawAII) Vertically Integrated Project (VIP), supervising students on vision-language-action models and embodied AI.

Meta AI Research 2022 - 2023
Research Scientist

- Ensured the resiliency to hardware errors of language model training experiments distributed over thousands of GPU machines.
- Developed a theory of instabilities in large language models' training quality.
- Reduced the training compute and the inference latency of a large transformer-based model, increasing the capacity of the language model's context by order of magnitude.

University Of California, Berkeley 2017 - 2022
Graduate Student Researcher

- Conducted research in non-convex numerical optimization with a team of 5 graduate students and faculty
- Developed the first conditions for guaranteed convergence of local search methods in structured non-convex matrix sensing that can be applied to real-world cases in electrical grid analysis
- Developed an algorithm for provably secure analysis of electrical grid data, which is robust to cyberattacks
- Theoretically explained the interpolation phenomenon in a popular machine learning algorithm MAML

- Advisor: Prof. Javad Lavaei

Facebook AI Foundations

Summer 2021

Machine Learning Engineering Intern

- Developed an on-device multilingual text classification model to enforce the integrity of ads in all of the Facebook applications around the world
- Achieved around 24% improvement in the target metrics over the in-production English-specific model retrained on a multilingual dataset
- For some policies, the proposed model leaves only 5 – 10% gap in the precision compared to the on-server model for multilingual classification (which is 1000 times larger in size)

Google DeepMind

Spring 2021

Research Scientist Intern

- Developed and tested a meta-learning approach for reinforcement learning agent modeling in a multiagent ad-hoc cooperative setting, solving tiny Bridge and tiny Hanabi games as milestones towards the Hanabi challenge
- Accelerated the proposed Meta-RL algorithm by a factor of 3 using offline reinforcement learning
- Advisors: Prof. Michael Bowling

Los Alamos National Lab

Summer 2017

Visiting Researcher

- Developed and implemented an accelerated algorithm for the solution of a nonconvex optimization problem in complex-valued power systems data analysis (Matlab+Python)
- Advisor: Prof. Michael Chertkov

Skolkovo Institute for Science and Technology (Moscow)

2016-2017

Research Intern

- Implemented novel numerical optimization technique; adapted the code for applications in large-scale electrical grid control, obtaining 30% gain in speed compared to the out-of-the-box solution
- Advisor: Dr. Yury Maximov

Pasteur Institute (Paris)

Summer 2016

Research Intern

- Implemented software (Java) and hardware (Arduino) solutions for a prototype of the technology of in-house hydrogel chip production for biomedical laboratories, reducing the design cycle from several weeks to several hours
- Advisor: Dr. Samy Gobaa

ADVISING AND MENTORSHIP

- **Ph.D. Students:** Yosub Shin (2024 - current, ECE/ICS/Robotics), Nathaniel Anleitner (ECE), Arnav Mehrotra (ECE), Pavel Bushuyeu (ICS), Andre Shiu (ECE).
- **Master's Students:** William Paek (Math), Shayla Sharmin Urmi (ECE), Michael Buriak (ICS), Kayla-Marie Torres (ICS).
- **Graduated Students:** Dr. Honggen Zhang (Ph.D. 2025, Dissertation: "LLM research and applications", now at XtalPi), Amila Indika (M.S. 2025, co-supervised, now in industry).

PUBLICATIONS

Preprints and Working Papers:

- A. Bozorgkhoo and I. Molybog, “Speculative Decoding Scaling Laws (SDSL): Throughput Optimization Made Simple,” *arXiv preprint arXiv:2603.11053*, 2026.
- A. Indika and I. Molybog, “SODBench: A Large Language Model Approach to Documenting Spreadsheet Operations,” *arXiv preprint arXiv:2510.19864*, 2025.
- Y. Shin and I. Molybog, “Beyond Audio and Pose: A General-Purpose Framework for Video Synchronization,” *arXiv preprint arXiv:2506.15937*, 2025.
- I. Molybog, “What if synthetic data is all you need,” *ResearchGate preprint dx.doi.org/10.13140/RG.2.2.11362.75209*, 2024.
- A. Dubey, A. Jauhri, A. Pandey *et al.* (including I. Molybog), “The Llama 3 Herd of Models,” *Meta AI Technical Report, arXiv:2407.21783*, 2024.
- I. Molybog, P. Albert, M. Chen *et al.*, “A Theory on Adam Instability in Large-Scale Machine Learning,” *arXiv preprint arXiv:2304.09871*, 2023.
- H. Touvron, L. Martin .. *et al.* (including I. Molybog), “Llama 2: Open Foundation and Fine-Tuned Chat Models,” *Meta AI Research Report*, 2023.

Conference Proceedings:

- Y. Shin, M. Buriek, B. Sobolev, P. Bushuyeu, V. Kumar, H. Xu, S. Watson and I. Molybog, “What Matters in Data Curation for Multimodal Reasoning? Insights from the DCVLR Challenge,” in *ICML Workshop on Efficient Multimodal Question Answering*, 2026.
- Y. Shin, M. Buriek and I. Molybog, “m2sv: A Scalable Benchmark for Map-to-Street-View Spatial Reasoning,” in *ICML Workshop on Combining Theory and Benchmarks*, 2026.
- Y. Wen, I. Molybog *et al.*, “Anti-Knowledge Corruption: Document Augmentation Defense,” in *Annual Computer Security Applications Conference (ACSAC)*, 2025.
- H. Zhang, X. Zhao, I. Molybog and J. Zhang, “REAL: Response Embedding-based Alignment for LLMs,” in *IJCAI Workshop on Causal Learning for Recommendation Systems*, 2025.
- W. Xiong, J. Liu, I. Molybog *et al.*, “Effective Long-Context Scaling of Foundation Models,” in *North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.
- H. Zhang, J. Zhang and I. Molybog, “HaSa: Hardness and Structure-Aware Contrastive Knowledge Graph Embedding,” in *The Web Conference (WWW)*, 2024.
- Z. Ma, I. Molybog, S. Sojoudi and J. Lavaei, “Over-parametrization via Lifting for Low-rank Matrix Sensing: Conversion of Spurious Solutions to Strict Saddle Points,” in *International Conference on Machine Learning (ICML)*, 2023.
- I. Molybog and J. Lavaei, “Global Convergence of MAML for LQR,” in *IEEE Conference on Control Technology and Applications (CCTA)*, 2021.
- I. Molybog and J. Lavaei, “On Sampling Complexity of the Semidefinite Affine Rank Feasibility Problem,” in *AAAI Conference on Artificial Intelligence*, 2019.
- I. Molybog and J. Lavaei, “Towards Robust and Scalable Power System State Estimation,” in *IEEE Conference on Decision and Control (CDC)*, 2019.
- I. Molybog and J. Lavaei, “Conic Optimization for Robust Quadratic Regression: Deterministic Bounds and Statistical Analysis,” in *IEEE Conference on Decision and Control (CDC)*, 2018.

Journal Papers:

- I. Molybog, R. Madani and J. Lavaei, “Conic Optimization for Quadratic Regression Under Sparse Noise,” *Journal of Machine Learning Research*, vol. 21, no. 195, pp. 1–37, 2020.
- I. Molybog, S. Sojoudi and J. Lavaei, “Role of Sparsity and Structure in the Optimization Landscape of Non-convex Matrix Sensing,” *Mathematical Programming*, vol. 182, no. 1, pp. 1–37, 2020.
- M. Jin, I. Molybog, R. Mohammadi-Ghazi and J. Lavaei, “Scalable and Robust State Estimation from Abundant Data,” *IEEE Transactions on Smart Grid*, vol. 11, no. 3, pp. 1880–1894, 2019.

Ph.D. Thesis:

- *I. Molybog*, “The complexity of non-convex and conic optimization problems in data science applications,” Ph.D. Dissertation, University of California, Berkeley, 2022.

SKILLS

- Rapid software/hardware prototyping: Python(Expert) / C/C++ / Arduino
- Cloud, distributed and parallel computing tools: GCP / AWS / Slurm / CUDA / NCCL / Apex / Triton
- Open-source development: pytorch(distributed) / fairscale / fairseq / metaseq / xformers
- Open-source tools: TensorFlow / Jax / pandas / scikit-learn

LEADERSHIP AND TEACHING EXPERIENCE

- Graduate Student Instructor (TA) at the University of California, Berkeley 2018 – 2021
Served as an Assistant for Optimization Analytics (IEOR 240), and Nonlinear and Discrete Optimization (IEOR 160) and instructor for Integer Optimization (IEOR 169).
- Industry advisor of the faculty hiring committee at the Department of Electrical and Computer Engineering of the University of Hawai’i at Manoa 2023
- The chair of a section at the INFORMS Annual Meeting 2021
Organised the section on Reaching Global Optimum in Non-Convex Optimization Problems
- Mentor on the First Steps in Research program at UC Berkeley 2021
- Co-founder of Align.ai 2019
Founded an AI-as-a-service startup in commercial real estate space, grew the team to five people and took it through the UC-based Launch Accelerator
- The president of the Russian Speaking Student Association at UC Berkeley 2019 - 2021
Curated social, networking and learning events for approximately 100 senior and graduate students
- Undergraduate Student-Mentor at Moscow Institute of Physics and Technology 2016 - 2017
Supervised two cohorts of students, more than 25 students each
- Member of the University Basketball team at MIPT 2013 - 2015